Sequences & Series

Section 1: General Arithmetic & Geometric Progressions

1 In a geometric sequence, the 2nd term is 6 and the 5th term is 162. What is the common ratio r?

a. 2

b. 3

c. 4

d. 9

2 Insert two geometric means between 5 and 320. What is the first inserted term?

a. 10

b. 20

c. 40

d. 80

3 Find the sum of all multiples of 4 between 10 and 100 inclusive.

a. $1,188$

b. $1,210$

c. $1,260$

d. $1,320$

Section 2: Sigma Notation & Special Sums

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4 Evaluate the sum: $\sum_{k=1}^{20} (3k + 2)$.

a. 650

b. 670

c. 690

d. 710

5 Using the formula $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$, evaluate $\sum_{k=1}^{10} k^2$.

a. 385

b. 420

c. 505

d. 550

6 What is the sum of the telescoping series $\sum_{n=1}^{50} (\frac{1}{n} - \frac{1}{n+1})$?

a. $\frac{49}{50}$

b. $\frac{50}{51}$

c. 1

d. $\frac{51}{52}$

Section 3: Infinite Series Convergence

7 For what values of x does the geometric series $\sum_{n=0}^{\infty} (2x - 3)^n$ converge?

a. $1 < x < 2$

b. $0 < x < 3$

c. $-1 < x < 1$

d. All real x

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8 What is the sum of the series $\sum_{n=1}^{\infty} \frac{3}{4^n}$?

a. $\frac{3}{4}$

b. 1

c. $\frac{4}{3}$

d. 2

9 Does the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ converge or diverge?

a. Converges to 1

b. Converges to $\pi^2/6$

c. Diverges

d. Oscillates

Section 4: Recursive Sequences & Induction

10 A sequence is defined recursively by $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. What is the value of a_4?

a. 9

b. 15

c. 17

d. 21

11 If $F_1 = 1, F_2 = 1,$ and $F_n = F_{n-1} + F_{n-2}$ represents the Fibonacci sequence, what is F_8?

a. 13

b. 21

c. 34

d. 55

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12 In a proof by mathematical induction, after verifying the base case $P(1)$, what is the inductive hypothesis?

a. Assume $P(k)$ is true for some integer $k \geq 1$

b. Prove $P(1)$ is false

c. Assume $P(k+1)$ is true

d. Prove for all k simultaneously

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